01057033 [illegible]

Ch 5_Quiz

Multiple Choice Questions (10%)

D (B) D 1. Which of the following is not a component of repetitive control? P.27、figure 5.7

A. modify B. test C. initialize D. recursion

2

B (A) B 2. What is known as an assertion at a point in a loop that is true every time that point in the loop is reached?

A. precondition B. loop invariant C. postcondition D. termination

每次回圈必定成立的事實：P.46

(D) D 3. The binary search algorithm is an example of an algorithm in which of the following classes?

A. $\Theta(n^2)$ B. $\Theta(n)$ C. $\Theta(n \log_2 n)$ D. $\Theta(\log_2 n)$

(B) B 4. When searching within the list (Lewis, Maurice, Nathan, Oliver, Pat, Quincy, Roger, Stan, Tom), which of the following entries will be the second one found using the sequential search algorithm?

A. Lewis B. Maurice C. Tom D. Pat

D (~~A~~ B) D 5. Which of the following is a loop invariant at the point at which the test for termination is performed in the following loop structure?

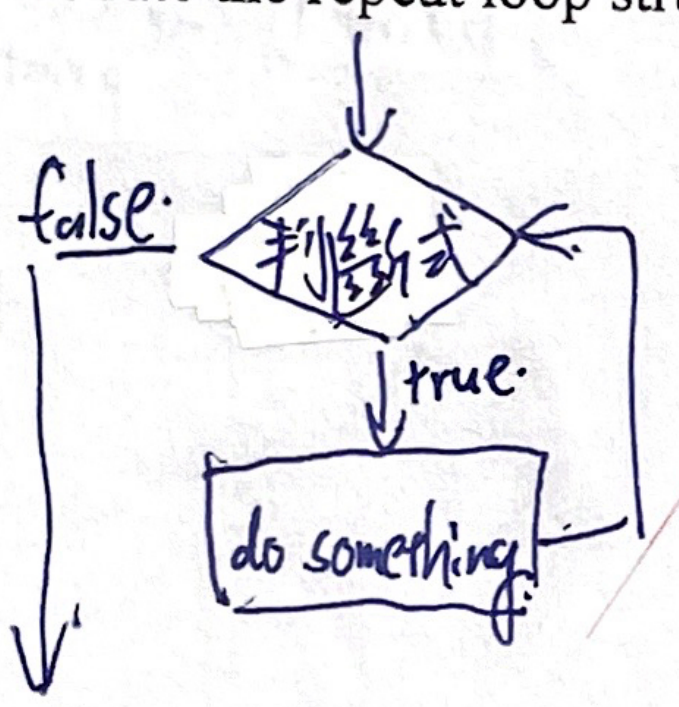
每次執行回圈必定成立的事實

```
X = 5
while (X < 10):
    X = X*2
```

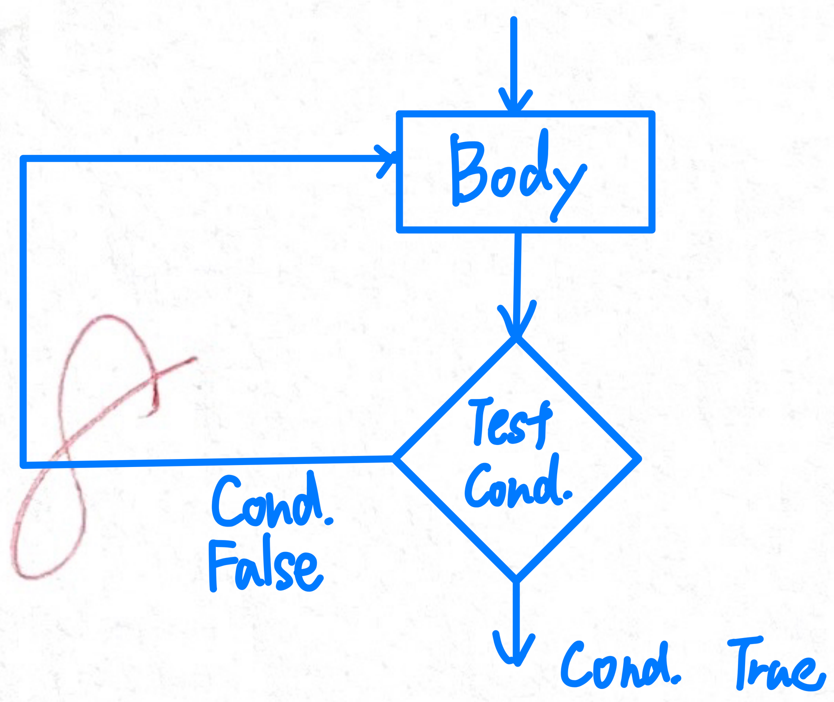
A. X > 5 B. X ≤ 5 C. X > 10 D. X ≤ 10

Fill-in-the-blank/Short-answer Questions (90%)

1. Draw a flow chart to illustrate the repeat loop structure. (8%)



-8



2. Fill in the blank in the function below so that the function prints 567753 (8%)

```
X = 5
while (X < 7):
    print(X)        → 5、6
    X = X + 1
print(X)            → 7
while ( X≥3 ):      → X≥3
    print(X)        → 7、5、3
    X = X - 2
```

3. At most, how many entries in a list of 11 names will be interrogated when using binary search algorithm? (8%)

$\log_2 11 \approx 3.X \Rightarrow 4$ $\log_2 11$ 4 -4

4. At most, how many comparisons in a list of 10 names will be made when using insertion sort algorithm? (8%)

0、1、2、……、9

$\frac{(0+9)\times 10}{2} = 45$ 次 100 45 -8

5. Which of the sequential or binary search algorithms would find the name Kelly in the list more quickly?. (8%)

liner 二分

John, Kelly, Lewis, Maurice, Nathan, Oliver, Pat, Quincy, Roger, Stan, Tom

Sequential. sequential

6. Suppose the binary search algorithm was being used to search for the entry Tom in the list

Nathan, Oliver, Pat, Rodger, Tom, Stan

A. What would be the first entry in the list to be considered? ~~Pat~~ Rodger (5%)

B. What would be the second entry in the list to be considered? Tom' Tom (5%)

-5

7. What sequence of numbers would be printed if the following function were executed with the value of N being 1? (8%)

```
def xxx(N):
    print(N)
    if (N < 2):
        xxx(N + 1)
    else:
        print(N)
    print(N + 1)
```

xxx 1 — print 1, xxx 2 — print 2, print 2, print 3; print 2

Ans. 1 2 2 3 2.

xxx(1) — print(1), xxx(2) — print(2), print(2), print(3); print(2)

A: 1 2 2 3 2

8. What would be printed if the value of $\boxed{\text{N being 5}}$? (8%)

```
def xxx(N):
    if (N > 0):
        xxx(N - 1)
    print(N)
```

xxx 5 — xxx 4 — xxx 3 — xxx 2 — xxx 1 — xxx 0; print 5, print 4, print 3, print 2, print 1, print 0

Ans 0 1 2 3 4 5

xxx(5) — xxx(4) — xxx(3) — xxx(2) — xxx(1) — xxx(0); print(5) print(4) print(3) print(2) print(1) print(0)

Ans: 0 1 2 3 4 5

9. Use a repeat loop rather than a while loop to accomplish the same results as the following program segment. (8%)

```
X = 1
while (X < 5):
    print(X)
    X = X + 1
```

for i in range(1,5):
    print(i)

Ans 1. 2. 3 4.

-8

10. Use a while loop structure to produce a non-recursive program segment that prints the same sequence of numbers as the following recursive function. (8%)

```
def xxx(N):
    print(N)
    if (N < 8):
        xxx(N + 1)
```

do:
    print(N)
    N+=1;
while(N<8)

while (True):
    print(N)
    N+=1
    if(N≥8):
        break.

while (N<8):
    print(N)
    N=N+1;

-8

11. What value would be printed if the following function were executed with the value of N being 4 and 9, separately? (8%)

```
def xxx(N):
    if (N < 7):
        print(N)
    else:
        N = N + 3
        print(N)
```

N=4. Ans: 4. N=9. Ans. 12.

N=4 Ans: 4 N=9 Ans: 12